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Garlic-specific Fertilizer Improves Economic and Environmental Outcomes in China

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Abstract. **Traditional methods of garlic fertilization involve large amounts of balanced fertilizer with equal proportions of N, P, and K, leading to nutrient imbalances, reduced yield and nutritional quality, and elevated risk of environmental pollution. This study for the first time measured garlic nutrient absorption and mineral elements status in garlic fields. In addition, a garlic-specific fertilizer formula and recommended rate were designed and applied in multiple garlic fields during the 2019–21 growing season. We assessed the performance of garlic-specific fertilizer in terms of yield, quality, and nutrient utilization efficiency. We showed that garlic prefers to absorb N and K, and its absorption of P was much lower. Deficiencies in Cl, Mn, S, and Fe are found in 98.7%, 56.1%, 22.8%, and 11.9% of garlic fields. Compared with farmer fertilization, the garlic-specific fertilizers increased sprout yield by 12.9% to 30.5%, bulb yield by 11.0% to 33.5%, and net income by 18.2% to 45.6%. Furthermore, it improved the nutritional quality [vitamin C (Vc), soluble sugar (SS), and soluble protein] of the garlic and reduced the accumulation of nitrate. The formula of special fertilizer was more in line with the law of garlic nutrient absorption, increasing the nutrient utilization effect, reducing the environmental risks. Application of specific fertilizer increased N, P, and K partial productivity by 26.6% to 50.1%, 82.6% to 116.5%, and 54.6% to 83.3%, respectively. These results suggest that replacing balanced fertilizers in the garlic market with garlic-specific fertilizers can improve garlic farmers' incomes and soil health.**

Garlic (*Allium sativum* L.) is the second most widely used allium vegetable after onion, typically as a spice or condiment (Sung et al. 2014). Moreover, garlic contains many minerals, vitamins, and allicin that have been shown to help treat cardiovascular diseases, stomach diseases, sore eyes, and earache (Mondal et al. 2022; Shang et al. 2019). Garlic is highly adaptable and extensively cultivated worldwide. On a global scale, the leading producers are China, India, Bangladesh, Korea, Egypt, and Spain. Garlic is predominantly grown in Asia (87%), where China and India collectively account for 78% of global production (Wang et al. 2022). According to the Food and Agricultural Organization (FAO) database, the global production of garlic reached 30.7 million tons with a land area of 1.6 million hectares in 2019. China produces 23.3 million tons of garlic annually with a cultivated area of 0.83 million hectares concentrated in four provinces: Shandong, Henan, Jiangsu, and Hebei (FAO 2019). Qi County, located in Henan Province, has cultivated garlic for more than 2000 years, ranking first among Chinese counties and known as the "hometown of garlic in China" (Zhang and Li 2021).

As an economic crop, garlic reaps greater benefits than wheat, maize, and other food crops. In pursuit of higher revenue, garlic farmers often invest a large amount of fertilizer to improve yields. However, blind fertilization is commonly practiced due to poor fertilization guidance. Excessive fertilization causes a large amount of mineral element residues in the soil, which results in nutrient leaching and volatilization, aggravating the soil compaction, acidification, and salinization (Toor et al. 2020). Besides excessive fertilization, local farmers are accustomed to using compound fertilizers with equal proportions of nitrogen (N), phosphorus (P), and potassium (K) (such as N–P–K = 18–18–18, 17–17–17, 15–15–15). Moreover, they do not consider organic and biological fertilizers or soil supplementation with trace elements. These oversights decrease fertilizer utilization efficiency for garlic cultivation; N, P, and K fertilizer utilization efficiency is only ∼35%, 10%, and 40%, respectively. Meanwhile, soil nutrient imbalance reduces soil quality and therefore garlic yield and quality while even risking disease (Norris and Congreves 2018). In recent years, the research on garlic fertilization has mainly focused on how to optimize the fertilization strategy to improve the yield and quality of garlic, and the application of organic fertilizer has gradually attracted attention, and further attention may be paid to more efficient and environmentally friendly fertilization strategies in the future (Li et al. 2018; Wang et al. 2022; Wei et al. 2023).

Large quantities of fertilizer and low utilization efficiency in garlic cultivation can also harm the environment. To protect the Erhai Lake, the government of Dali, a county in Yunnan Province of China, has banned farmers from planting garlic (Duan et al. 2021). Therefore, this study aims to design a specialized fertilizer suitable for regional garlic cultivation that increases garlic yield and quality as well as fertilizer utilization efficiency while reducing environmental pollution.

Materials and Methods

Site description

Nitrogen, phosphorus, and potassium rate field experiments were conducted in Jiangzhai (34°41′N, 114°39′E) and Caotun village (34°31′N, 114°49′E), in Qi county (garlic planting area ≥ 4.5 × 10^4 ha/year; garlic yield ≥ 90 × 10^4 t/year), Henan Province, Central China, during the Oct 2018 to May 2019 garlic growing season. The area has a subtropical monsoon climate with a mean annual temperature of 9.6 °C; the mean winter temperature was 1.5 °C during the garlic growing season. The total rainfall during the garlic growing season was 361 mm. Before initiating the experiment, soil samples were extracted from the upper 20-cm layer for chemical analyses. Table 1 lists the physicochemical properties of the soil.

Special fertilizer field experiments were conducted in Xiaogang (34°29′N, 114°43′E), Jiangzhai (34°40′N, 114°41′E), Xizhai (34°37′N, 114°52′E), and Chenlou (34°30′N, 114°52′E) village, in Kaifeng City, Henan Province, during the Oct 2019 to May 2020 garlic growing

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Table 1. Basic soil (0–20 cm) physicochemical properties.

Yr	Site	pH	Soil bulk density ($g \cdot cm^{-3}$)	Organic matter ($g \cdot kg^{-1}$)	Total nitrogen ($g \cdot kg^{-1}$)	Available phosphorus ($mg \cdot kg^{-1}$)	Available potassium ($mg \cdot kg^{-1}$)
2018–19	Caotun	6.6	1.34	13.2	1.8	22.2	147.5
	Jiangzhai	6.6	1.23	8.6	0.9	18.2	69.5
2019–20	Xiaogang	7.1	1.28	10.8	1.1	18.4	106.3
	Jiangzhai	6.2	1.26	8.9	0.8	19.6	87.5
	Xizhai	6.8	1.31	9.1	0.9	17.2	77.0
	Chenlou	7.2	1.24	15.6	1.6	20.4	127.3
2020–21	Caotun	6.8	1.34	12.1	1.4	21.4	103.9
	Jiangzhai	6.6	1.23	8.8	0.9	16.5	72.4
	Mamiao	7.4	1.31	12.2	1.3	15.5	122.6
	Zuozhai	7.3	1.29	7.6	0.8	15.8	73.7
	Wanghesi	7.2	1.30	8.9	1.0	16.5	98.1
	Chenlou	7.2	1.24	10.3	1.2	17.4	109.0

season; from Oct 2020 to May 2021, the same experiments were conducted in Caotun (34°32′N, 114°46′E), Jiangzhai (34°40′N, 114°38′E), Mamiao (34°30′N, 114°38′E), Zuozhai (34°25′N, 114°51′E), Wanghesi (34°28′N, 114°54′E), and Chenlou (34°30′N, 114°49′E) villages. The detailed distribution of experiment locations is shown in Fig. 1. The mean annual temperatures were 10.6 °C (2019–20) and 9.4 °C (2020–21), whereas the mean winter temperatures were 1.5 °C (2019–20) and 1.1 °C (2020–21) during the garlic growing season. The total rainfall during the garlic growing season was 310 mm (2019–20) and 385 mm (2020–21).

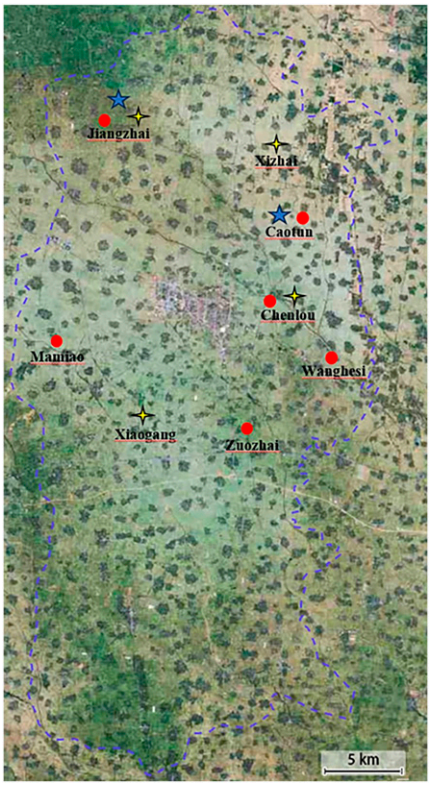

Fig. 1. Distribution of experiment locations. The experiment was carried out at Qixian County, Kaifeng City, Henan Province (34.23°–34.76°N, 114.61°–114.94°E) from 2018 to 2021. Nitrogen, phosphorus, and potassium rate field experiments were conducted at Jiangzhai and Caotun villages [presented as (★)], during the Oct 2018 to May 2019 garlic growing season. Special fertilizer experiments were conducted at Xiaogang, Jiangzhai, Xizhai, and Chenlou villages [presented as (✦)], during the Oct 2019 to May 2020 garlic growing season; from Oct 2020 to May 2021, the same special fertilizer experiments were conducted at Caotun, Jiangzhai, Mamiao, Zuozhai, Wanghesi, and Chenlou villages [presented as (●)].

Experimental design and management

Questionnaire survey. A baseline household survey on garlic management practices was administered to farmers from 100 households who worked in the sampled fields in Jul and Aug 2020. All in-house surveys were conducted by trained investigators. Each respondent verbally consented before beginning the survey.

N, P, and K fertilizer rate experiment. The same cultivar (Jinan Daqingke) treatments and crop management strategies were used at both study sites. A total of 416,880 seeds/ha were manually sown on 7 Oct. The dimensions of each plot were 6.0 m × 8.0 m. The N fertilizer rate study was arranged in a randomized complete block with three replicates, and five N base fertilizer treatments were used: 0, 150, 300, 450, and 600 $kg \cdot ha^{-1}$ N. N in the form of urea (46% N), P (200 $kg \cdot ha^{-1}$ P_2O_5) in the form of calcium superphosphate (12% P_2O_5), and K (200 $kg \cdot ha^{-1}$ K_2O) in the form of potassium chloride (60% K_2O) were applied as the basal dose.

The P fertilizer rate study was arranged in a randomized complete block with three replicates, and five P base fertilizer treatments were used: 0, 75, 150, 225, and 300 $kg \cdot ha^{-1}$ N. N (300 $kg \cdot ha^{-1}$ N) in the form of urea (46% N), P in the form of calcium superphosphate, and potassium (200 $kg \cdot ha^{-1}$ K_2O) in the form of potassium chloride (60% K_2O) were applied as the basal dose.

The K fertilizer rate study was arranged in a randomized complete block with three replicates, and five K base fertilizer treatments were used: 0, 75, 150, 225, and 300 $kg \cdot ha^{-1}$ (N). N (300 $kg \cdot ha^{-1}$ N) in the form of urea (46% N), P (200 $kg \cdot ha^{-1}$ P_2O_5) in the form of calcium superphosphate (12% P_2O_5), and potassium in the form of potassium chloride (60% K_2O) were applied as the basal dose.

The N, P, and K fertilizer rate experiments were initiated during bulb generation (≈150 d after sowing) by a topdressing fertilizer including 225 $kg \cdot ha^{-1}$ urea and 75 $kg \cdot ha^{-1}$ KH_2PO_4, followed by a second application of 150 $kg \cdot ha^{-1}$ KH_2PO_4 during bulb expansion (≈180 d after sowing). Weeds and insects were closely monitored to minimize yield losses.

Special fertilizer field experiments. The same cultivar (Jinan Daqingke) treatments and crop management strategies were adopted across study sites during the two growing seasons. Two treatments were used in the experiment: farmer practice (FP) and garlic-specific fertilizer (SF). Local farmers are accustomed to using 1800 $kg \cdot ha^{-1}$ pure sulfur-based compound fertilizer as the base fertilizer (18–18–18). The garlic-SF was a low chlorine compound fertilizer (18–10–14) supplemented with microelements (S, Mn, Fe, and Zn) produced by Henan Xinlianxin Chemical Industry Group Co., Ltd and applied at a rate of 1500 $kg \cdot ha^{-1}$. The plot area was 1200 m^2 with a row spacing of 20 cm and a plant spacing of 12 cm. The first topdressing fertilizer was applied during bulb degeneration (≈150 d after sowing) with 225 $kg \cdot ha^{-1}$ urea and 75 $kg \cdot ha^{-1}$ KH_2PO_4, followed by a second application of 150 $kg \cdot ha^{-1}$ KH_2PO_4 during bulb expansion (≈180 d after sowing). Detailed fertilization information is shown in Table 2.

Plant sampling and measurement of parameters

Soil mineral element measurement. Before garlic planting, we collected soil from 10 fields from each of 54 garlic planting villages in Qixian County, Kaifeng, Henan Province. Each soil sample was placed on a sample tray, spread into a thin layer, and dried in a ventilated indoor space without exposure to sunlight or contaminants like acids and alkalis. After air

Table 2. Nutrient input per hectare for farmer practice and special fertilizer treatment. During the 2019–20 growing season, the special fertilizer experiment was arranged at Xiaogang, Jiangzhai, Xizhai, and Chenlou sites; during the 2020–21 growing season, the experiment was arranged at Caotun, Jiangzhai, Mamiao, Zuozhai, Wanghesi, and Chenlou sites.

Treatment	Base fertilizer	First topdressing fertilizer	Second topdressing fertilizer	N ($kg \cdot ha^{-1}$)	P_2O_5 ($kg \cdot ha^{-1}$)	K_2O ($kg \cdot ha^{-1}$)
Farmer practice	Compound fertilizer 18–18–18 1800 $kg \cdot ha^{-1}$	Urea 225 $kg \cdot ha^{-1}$, KH_2PO_4 75 $kg \cdot ha^{-1}$	KH_2PO_4 150 $kg \cdot ha^{-1}$	427.5	441	400.5
Special fertilizer	Special fertilizer 18–10–14 1500 $kg \cdot ha^{-1}$			373.5	267	286.5

drying, the sample was laid flat on the sample plate, crushed with a wooden stick, and cleared of plant residues and stones. The crushed soil samples were filtered through a 2-mm nylon mesh sieve, during which a part of the sample was removed by the quartering method and ground so that it could pass through a 0.25-mm nylon sieve. Another part of the sample that passed through the 0.25-mm nylon sieve was removed by quarrying and was ground with an agate mortar to pass through a nylon sieve with 0.149-mm pores. Processed samples were stored in plastic bottles for later use.

After the soil samples were digested with acid, their total N content was determined by an automatic N determination instrument. The alkaline hydrolyzable N content of the soil samples was determined by titration with $1/2H_2SO_4$ after alkaline diffusion with NaOH. After the soil samples were extracted with $NaCO_3$, molybdenum-antimony anticolorimetric agent was added to determine the amount of available potassium at a wavelength of 700 nm using a spectrophotometer ultraviolet-5100 (Metash, Shanghai, China). Samples were extracted with Na_4OAc, and the available potassium content was determined by a flame photometer (FP-640; Precision Instrument Co., Ltd., Shanghai, China). Inductively coupled plasma mass spectrometry (ICP-MS) (7900; Agilent, Singapore) was used to quantify trace elements (Ca, Mg, S, Fe, Mn, Cu, Zn, B) in soil. The amount of water-soluble chlorine in soil was measured using $AgNO_3$ titration.

Plant nutrient measurement. Ten plants from the double row in each plot were sampled at the sowing stage (0 d after sowing; DAS), wintering stage (90 DAS), bulb differentiation stage (180 DAS), flowering stem elongation period (210 DAS), and bulb expansion period (240 DAS). After dissection into stem and leaf, bulb, and sprout, the fresh material was oven dried at 105 °C for 30 min and then at 75 °C until its weight stabilized. The materials were ground to pass through a 1-mm mesh screen and digested by H_2SO_4 and H_2O_2. The total N and P concentrations in the digested samples were determined using an automated continuous flow analyzer (Seal, Norderstedt, Germany). Total potassium (K) concentration of each plant was measured with a flame photometer (FP-640; Precision Instrument Co., Ltd.). Before quantifying medium- and microelements (Ca, Mg, S, Fe, Mn, Cu, Zn, B), plant materials were digested by HNO_3 and $HClO_4$. After constant volume diafiltration of the solution, various minerals in the test solution were measured using ICP-MS (7900; Agilent). Plant samples were extracted with water, and their Cl content was measured using an automated continuous flow analyzer (Seal, Norderstedt, Germany).

Nutritional quality measurements. At the harvest stage, avoiding the plot boundary (considering the marginal effect, the three plants near the border line are not selected), a total of 50 continuous plants from each plot were sampled to measure bulb yields. The bulb samples were used for assessing the nutritional quality (Vc, SS, soluble protein, and nitrate). The Vc concentration was determined as described by Danbature et al. (2015). The extracted sample was mixed with an ethanol solution (5% trichloroacetic acid, 0.4% H_3PO_4, 0.5% bathophenanthroline, 0.03% $FeCl_3$), and incubated at 30 °C for 60 min; its absorbance was measured at 534 nm.

SS content was measured as described by Zhang et al. (2020). The sample was mixed with 0.15% anthrone in ethanol and then incubated for 15 min in a water bath at 90 °C. The absorbance was measured at 620 nm using a PRIM light spectrophotometer (ultraviolet-1206; SHIMADZU, Japan). The SS content was determined from a standard curve of gradient D-glucose solutions.

Soluble protein content was measured as previously described in Bradford (1976). The

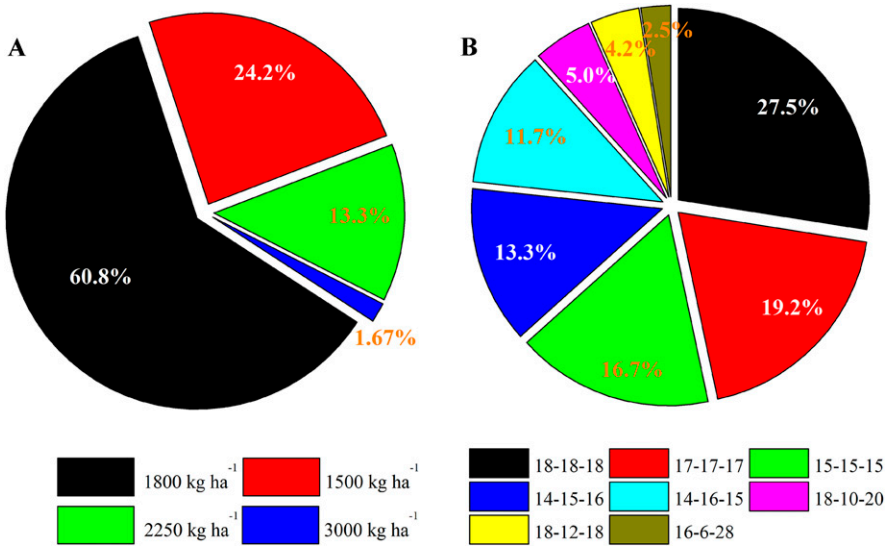

Fig. 2. Current fertilizer rate (**A**) and formula (**B**) among garlic farmers.

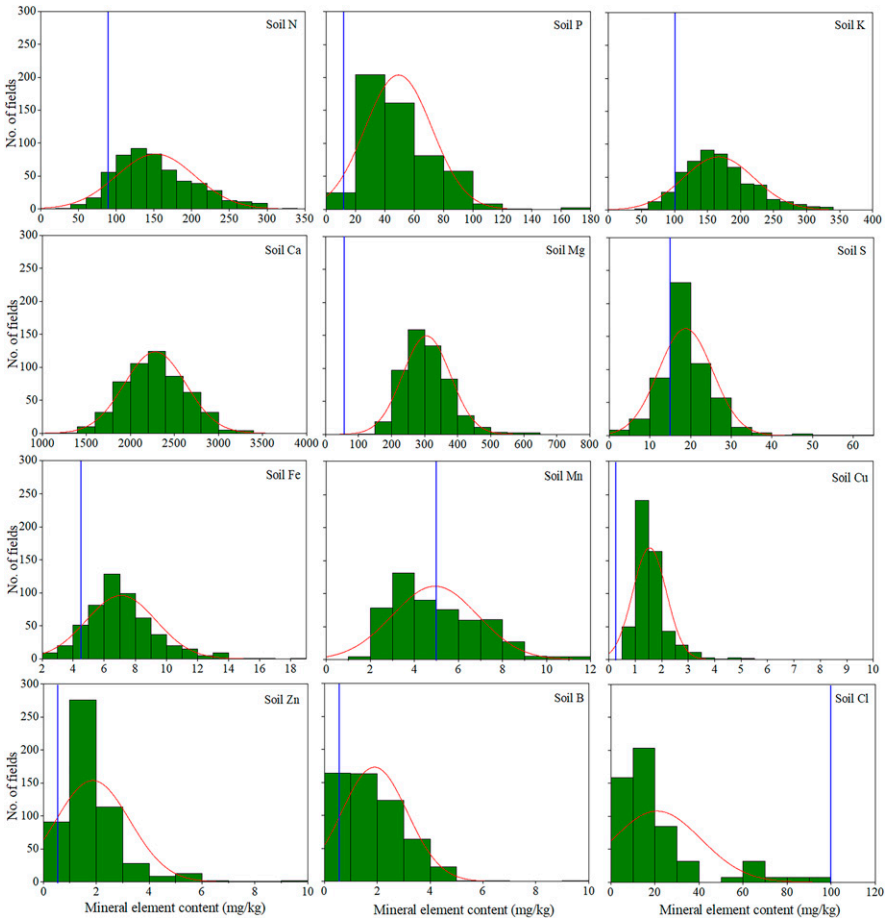

Fig. 3. Distribution of mineral elements in garlic fields in Qi county.

samples were homogenized with 25 mL H_2O for 30 min and centrifuged at 4000 g_n for 10 min. Extracted protein was stained using Coomassie Brilliant Blue G-250 and quantified via a spectrophotometer at a wavelength of 464 nm.

Nitrate content was measured as described by Singh et al. (2019). The homogenate was immersed in double-distilled water, and 5% salicylic acid in a sulfuric acid solution was added before incubation for 20 min at room temperature and addition of 1.8% sodium hydroxide. The absorbance was measured at 410 nm, and the nitrate level of all samples was calculated based on standard curves.

Data analysis

The fertilizer partial productivity parameters of the three nutrients (N, P, and K) were calculated as follows:

NPFP [kg·kg^{-1}] = garlic yield [kg·ha^{-1}]/N application rate [kg·ha^{-1}],

PPFP [kg·kg^{-1}] = garlic yield [kg·ha^{-1}]/N application rate [kg·ha^{-1}],

KPFP [kg·kg^{-1}] = garlic yield [kg·ha^{-1}]/N application rate [kg·ha^{-1}],

where NPFP is the N partial factor productivity, PPFP is the P partial factor productivity, and KPFP is the K partial factor productivity.

A one-way analysis of variance was performed to assess the differences within each parameter using the Statistical Software Package for Social Science (version 20.0). The mean values of the treatments were compared using the least significant difference test, where significance was $P < 0.05$. In the linear-plus-plateau regression model, the highest garlic bulb yield was obtained for a given N, P, and K fertilizer rate. All graphs were plotted using Origin 9.0 software.

Results

Farmer fertilization survey

According to the survey, 60.8% of farmers apply 1800 kg·ha^{-1} of the most commonly applied fertilizer on their garlic field (Fig. 2A). The least applied was 1500 kg·ha^{-1} of compound fertilizer, accounting for 24.2% of farmers. About 15% farmers applied more than 1800 kg·ha^{-1} fertilizers, and 1.7% applied 3000 kg·ha^{-1} fertilizers.

Farmers rarely considered the fertilizer formulations, and 63.4% of farmers chose a compound fertilizer with equal proportions of N, P, and K (Fig. 2B). Some businesses have manufactured similar fertilizer formulations such as 14–15–16/14–16–15, although they performed on par with 15–15–15 compound fertilizer. Fertilizer with unequal proportions of N, P, and K were used by fewer than 12% of farmers in Qixian County, where the two reasonable fertilizer formulas (18–10–20, 18–12–18) accounted for less than 10%.

Distribution of mineral elements in garlic fields

The distribution of mineral elements in the sampled garlic fields is shown in Fig. 3. Soil-available Ca, Mg, and Cu in garlic fields all exceeded the minimum standard. Nutrient deficiencies in N, P, K, Zn, and B were found in less than 10% of the fields, especially P (0.2%) and Zn (0.6%). Deficiencies in S, Fe, Mn, and Cl in soil were relatively high, reaching 22.2%, 11.9%, 56.1%, and 98.7%, respectively.

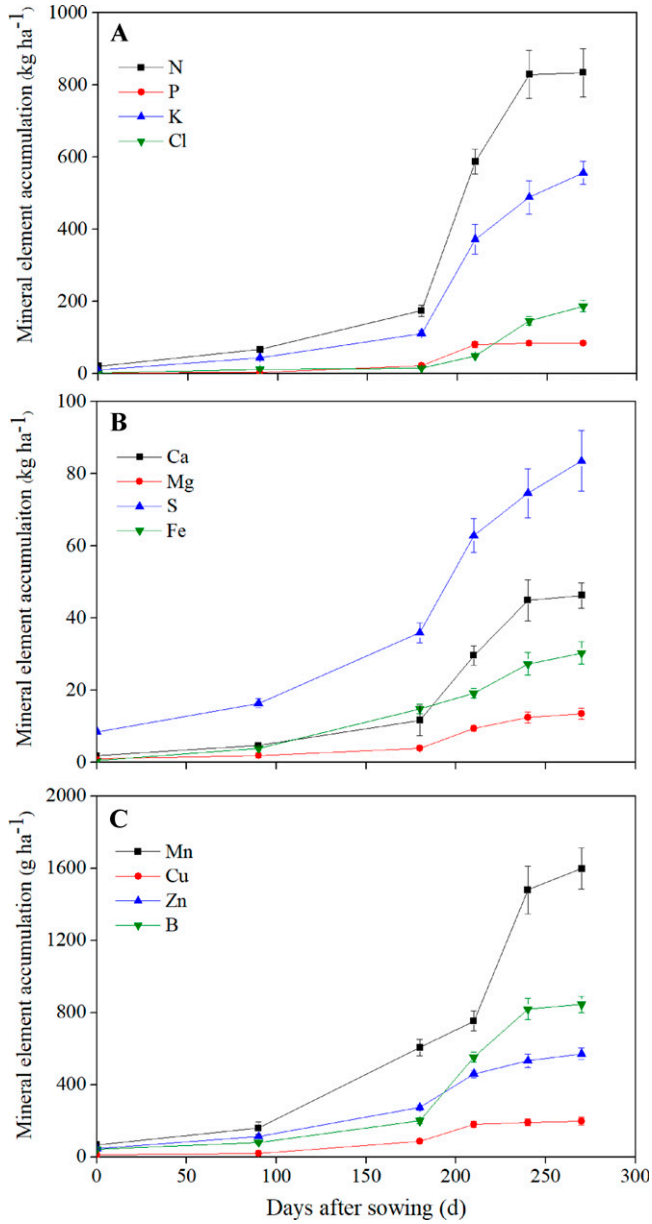

Fig. 4. Garlic absorption of macro- (**A**), medium- (**B**), and trace (**C**) elements. Each bar represents the mean ± *SD* (n = 3).

Nutrient uptake in garlic plants

Nutrient uptake in garlic significantly differed across mineral elements, which were divided into three groups based on the total amount absorbed by garlic plants: (1) macro-element group (mineral element accumulation >100 kg·ha^{-1}): N, P, K, and Cl; (2) medium-element group (10 kg·ha^{-1} < mineral element accumulation < 100 kg·ha^{-1}): Ca, Mg, S, and Fe; (3) micro-element group (mineral element accumulation < 10 kg·ha^{-1}): Mn, Cu, Zn, and B (Fig. 4).

The element most absorbed by garlic plants was N, followed by K. Notably, the absorption of Cl exceeded that of P. The garlic plants no longer absorbed P and Cu after 200 d and N, Ca, Mg, B, and Zn after 240 d. After 240 d, the plants absorbed K, Cl, S, Fe, and Mn.

Determination of the rate of N, P, and K fertilizer

In a certain range, bulb yield increased rapidly with the increase of N, P, and K fertilizers (Fig. 5). After reaching a certain amount of fertilization, the bulb yield no longer increased and even decreased thereafter. Using a linear-plus-plateau regression, we concluded that 272.4 kg·ha^{-1} was the highest yield application of N fertilizer, 154.2 kg·ha^{-1} was the highest yield application of P fertilizer, and 187.8 kg·ha^{-1} was the highest yield application of K fertilizer.

Effect of special fertilizer on the yield and quality of garlic

Yield and economic benefit. Application of SF affected bulb yield more than sprouts.

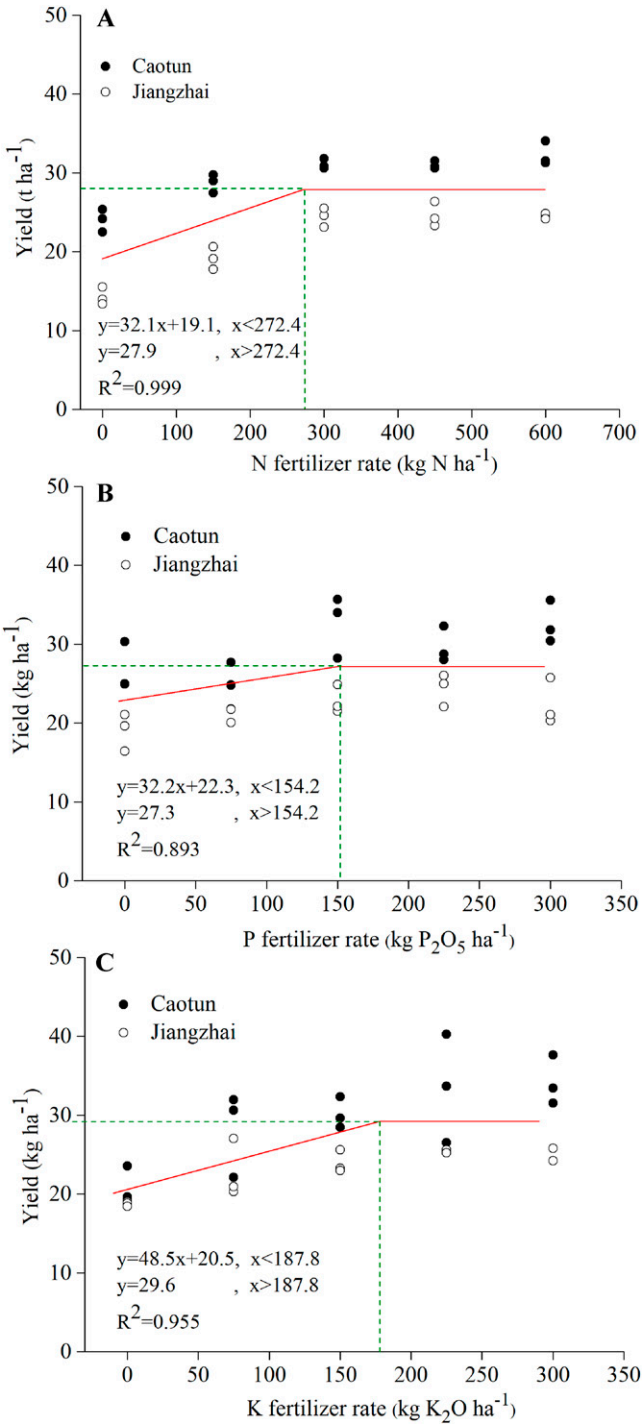

Fig. 5. Relationships between garlic yield and nitrogen (**A**), phosphorus (**B**), and potassium (**C**) fertilizer rates.

Compared with the FP treatment, the application of SF increased sprout yield by 6.2% to 24.6% (mean = 12.9%) and bulb yield by 2.7% to 22.2% (mean = 11.0%) during the 2019–20 growing season; the application of SF increased sprout yield by 23.5% to 40.4% (mean = 30.5%) and bulb yield by 20.2% to 47.5% (mean = 33.5%) during the 2020–21 growing season (Table 3). The price of sprouts remained relatively stable across seasons, whereas the price of bulbs varied greatly. Compared with the FP treatment, the application of SF increased output value by 2.8% to 22.4% (mean = 11.2%) and net income by 5.4% to 37.7% (mean = 18.2%) during the 2019–20 growing season; the application of SF increased output value by 20.2% to 47.5% (mean = 32.0%) and net income by 28.3% to 71.6% (mean = 45.6%) during the 2020–21 growing season.

Garlic quality. The application of specialized fertilizers improved not only the yield of garlic sprouts and bulbs but also their quality (Fig. 6). Compared with the FP treatment, the application of SF increased Vc content by 0.5% to 18.7% (mean = 9.5%), SS content by 32.0% to 38.3% (mean = 34.8%), and soluble protein content by 1.9% to 19.3% (mean = 11.3%) during the 2019–20 growing season; the application of SF increased Vc content by 21.2% to 138.8% (mean = 62.2%), SS content by 10.6% to 88.7% (mean = 49.9%), and soluble protein content by 1.6% to 47.7% (mean = 32.0%) during the 2020–21 growing season (Table 4).

Nitrate content was an important indicator of the quality of vegetables and was significantly reduced by scientific fertilization. In comparison with FP treatment, the application of SF reduced nitrate content by 35.4% to 46.7% (mean = 41.6%) during the 2019–20 growing season and by 13.6% to 47.7% (mean = 30.8%) during the 2020–21 growing season.

Fertilizer partial factor productivity. The application of SF enhanced the N fertilizer partial productivity of garlic by 26.6% and 50.1% during 2019–20 and 2020–21, respectively. The P fertilizer utilization efficiency improved the most, increasing by 82.6% and 116.5%, respectively. Second to P fertilizers was the K fertilizer utilization efficiency, as it improved by 54.6% and 83.3% in these two growing seasons, respectively (Fig. 7).

Discussion

This study characterized the nutrient absorption patterns of garlic. We found that garlic preferentially absorbs N and K, and its absorption of P was lower than that of Cl (Fig. 4A). Farmers prefer high nutrient concentrations of fertilizer and do not care about the formula. Currently, fertilizer companies provide garlic farmers with balanced fertilizers with consistent NPK ratios, such as 15–15–15, 17–17–17, and 18–18–18 because how garlic absorbs various nutrients remains unclear. Some fertilizer companies introduced 14–15–16 and 14–16–15 formulas, which among other balanced fertilizers comprise 88.3% of the garlic fertilizer market (Fig. 2B). Balanced fertilizers introduce a large amount of P into the soil, but garlic was not shown to have a high demand for P, resulting in excess P in the garlic soil (Fig. 3; Supplementary Table 1). Excessive P enhances plant respiration, which demands large amounts of sugar and energy; this thickens and darkens leaves while reducing the rate of photosynthesis (Malhotra et al. 2018; Sun et al. 2022). In addition, excessive P content can induce nutrient deficiencies in Fe, Zn, and Mg, as the combination of phosphate and metal ions forms insoluble substances in the soil (Osman and Osman 2013; Sharma et al. 2013).

The economic value of garlic was much higher than that of wheat; the net income of garlic farmers was 10.1×10^3 dollar/ha from 2019–20 and 12.5×10^3 dollar/ha from 2020–21, which were much higher than the 1.8×10^3 dollar/ha earned by wheat farmers (Table 3). Farmers are willing to invest in agricultural inputs, especially fertilizers, to improve yields. Today, more than 50% of garlic farmers believe that garlic yields increase

Table 3. Effect of special fertilizer on the garlic yield and farmer incomes. Garlic bulbs were sold for 0.42 dollar/kg in 2020 and 0.56 dollar/kg in 2020, and garlic sprouts were sold for 0.36 dollar/kg in 2020 and 0.42 dollar/kg in 2021. The price of 18:18:18 compound fertilizer used for farmer practice (FP) was 529 dollar/t, the price of 18:10:14 formula fertilizer used for special fertilizer (SF) was 557 dollar/t, and the prices of urea and KH_2PO_4 used for topdressing fertilizer were 362 and 1225 dollar/t, respectively. Other costs including seeds, pesticides, mulch, irrigation, and labor were calculated as 3.13×10^3 dollar/ha.

Site	Treatment	Sprout yield ($t \cdot ha^{-1}$)	Bulb yield ($t \cdot ha^{-1}$)	Output value ($\times 10^3$ dollar/ha)	Net income ($\times 10^3$ dollar/ha)
			2019–20		
Xiaogang	FP	3.22 ± 0.34 a	30.9 ± 2.0 b	14.1	9.6
	SF	3.57 ± 0.24 a	33.9 ± 2.1 a	15.5	11.1
Jiangzhai	FP	2.48 ± 0.13 b	26.1 ± 1.0 b	11.8	7.4
	SF	3.09 ± 0.27 a	31.9 ± 0.7 a	14.5	10.2
Xizhai	FP	2.89 ± 0.18 a	32.8 ± 0.6 a	14.8	10.3
	SF	3.07 ± 0.22 a	33.7 ± 1.7 a	15.2	10.9
Chenlou	FP	3.36 ± 0.27 a	39.0 ± 1.3 b	17.6	13.1
	SF	3.69 ± 0.36 a	42.7 ± 1.2 a	19.2	14.9
			2020–21		
Caotun	FP	3.66 ± 0.38 b	31.4 ± 2.1 b	19.1	14.6
	SF	4.52 ± 0.23 a	41.2 ± 2.9 a	23.7	19.4
Jiangzhai	FP	2.57 ± 0.35 b	26.0 ± 3.8 b	15.6	11.1
	SF	3.50 ± 0.12 a	34.8 ± 0.5 a	20.9	16.6
Mamiao	FP	2.86 ± 0.22 b	34.7 ± 0.9 b	20.6	16.2
	SF	3.54 ± 0.34 a	43.9 ± 3.4 a	25.9	21.6
Zuozhai	FP	2.83 ± 0.30 b	22.7 ± 2.3 b	13.8	9.3
	SF	3.61 ± 0.12 a	33.9 ± 2.3 a	20.3	16.0
Wanghesi	FP	2.96 ± 0.22 b	25.3 ± 4.0 b	15.3	10.9
	SF	3.90 ± 0.19 a	35.8 ± 2.7 a	21.6	17.3
Chenlou	FP	3.17 ± 0.20 b	28.6 ± 2.4 b	17.3	12.8
	SF	4.45 ± 0.24 a	33.9 ± 1.9 a	20.8	16.4

Different letters in a column denote significant differences between treatments at the same experimental site in the same year ($P < 0.05$).

with fertilizer input. Although we recommend a compound fertilizer dosage of 1500 $kg \cdot ha^{-1}$, only 24.2% of farmers have followed it; 60.8% of farmers exceed the fertilization limit by 20%, 13.3% of farmers exceed the fertilization limit by 50%, and 1.67% of farmers exceed it by 100% (Fig. 2A).

The profit of sulfur-based fertilizers was 34.8 dollar/t higher than that of chlorine-based fertilizers. Many argue that garlic is similar to tobacco, strawberry, and potato, is a Cl-sensitive crop, and can be used only with sulfur-based fertilizers. However, this study showed that garlic has a very high demand for Cl, and no references mentioned its sensitivity to chlorine (Fig. 4A). Long-term nonapplication of chloride-based fertilizers resulted in a significant deficiency of chlorine in the garlic field (Fig. 3), which significantly reduced the yield of garlic. Previous studies found that Cl can promote leaf photosynthesis and regulate stomatal movement (Franco-Navarro et al. 2019). During chlorine deficiency, leaf area and biomass significantly decreased (Zhao et al. 2005). Moreover, chlorinated fertilizers can inhibit plant diseases, such as wheat root rot, corn stem rot, and potato brown heart disease (Nadeem et al. 2018; Sharma et al. 2022; Singh 2015).

Table 4. Effect of special fertilizer on garlic bulb quality.

Site	Treatment	Vc (mg/100 g)	Soluble sugar (%)	Soluble protein (mg/g)	Nitrate (mg/kg)
			2019–20		
Xiaogang	FP	18.8 ± 0.2 a	8.1 ± 0.3 b	16.0 ± 0.1 a	996.8 ± 70.1 a
	SF	18.9 ± 1.9 a	11.2 ± 0.7 a	16.3 ± 1.0 a	531.6 ± 88.6 b
Jiangzhai	FP	21.9 ± 0.4 b	9.8 ± 0.4 b	18.7 ± 1.7 b	841.7 ± 38.8 a
	SF	26.0 ± 1.9 a	13.0 ± 0.4 a	22.3 ± 1.1 a	509.5 ± 44.7 b
Xizhai	FP	17.2 ± 1.9 a	7.5 ± 0.3 b	14.5 ± 0.2 b	886.0 ± 42.2 a
	SF	19.2 ± 0.6 a	9.9 ± 0.4 a	16.4 ± 0.6 a	487.3 ± 36.8 b
Chenlou	FP	19.8 ± 0.3 a	8.0 ± 0.4 b	16.3 ± 0.2 b	1063.2 ± 107.1 a
	SF	21.2 ± 1.4 a	10.9 ± 0.6 a	18.1 ± 0.6 a	686.7 ± 169.7 b
			2020–21		
Caotun	FP	11.8 ± 1.3 b	5.3 ± 1.1 b	15.6 ± 0.9 b	454.7 ± 38.0 a
	SF	14.3 ± 0.7 a	10.0 ± 1.4 a	18.3 ± 1.1 a	266.0 ± 27.3 b
Jiangzhai	FP	8.0 ± 1.1 b	4.8 ± 0.2 b	10.5 ± 0.6 b	605.4 ± 24.4 a
	SF	17.9 ± 3.3 a	6.8 ± 1.0 a	15.4 ± 1.6 a	523.0 ± 23.1 b
Mamiao	FP	8.5 ± 1.1 b	5.2 ± 1.1 b	15.1 ± 0.9 b	401.5 ± 41.0 a
	SF	20.3 ± 1.6 a	9.8 ± 1.4 a	22.3 ± 1.2 a	335.7 ± 29.9 b
Zuozhai	FP	10.6 ± 1.7 b	4.8 ± 0.2 b	12.7 ± 1.3 a	579.1 ± 36.7 a
	SF	14.4 ± 1.5 a	6.4 ± 1.3 a	12.9 ± 0.7 a	419.7 ± 83.4 b
Wanghesi	FP	12.0 ± 0.9 b	6.5 ± 0.8 b	8.6 ± 0.5 b	612.2 ± 80.9 a
	SF	15.7 ± 1.4 a	8.9 ± 0.7 a	12.0 ± 1.7 a	379.3 ± 34.1 b
Chenlou	FP	10.0 ± 1.2 b	6.6 ± 0.3 b	13.2 ± 1.1 b	542.1 ± 36.9 a
	SF	12.3 ± 0.5 a	7.3 ± 0.3 a	18.4 ± 1.9 a	283.5 ± 21.2 b

Different letters in a column denote significant differences between treatments at the same experimental site in the same year ($P < 0.05$). FP = farmer practice; SF = special fertilizer.

Long-term excessive application of balanced fertilizer tended to cause nutrient imbalances where minerals like N, P, K, Ca, Mg, Cu, Zn, and B were rich but others (Cl, Mn, S, and Fe) were lacking in the soil. This imbalance has largely contributed to the garlic continuous cropping obstacle (Tan et al. 2021). Excessive N fertilizer input ensured the high yield of garlic, once it encountered low temperature or higher soil moisture, it will greatly increase the probability of lateral shoot growth, resulting in substantial reduction or even no yield (Teixeira de Oliveira et al. 2020a, 2020b). A surplus of P and K is more suitable for the growth of pathogenic fungi, making plants more susceptible to disease (Zhao et al. 2021). In addition, deficiencies in Mn, Cl, and S can reduce plant immunity, as these minerals can protect against infectious plant diseases caused by pathogens such as fungi, bacteria, and viruses (Dordas 2008; Nadeem et al. 2018; Siddiqui et al. 2015).

We determined the N, P, and K fertilizer rates for garlic through elemental fertilizer field experiments. Some medium- and microelements (S, Cl, Mn, Fe, Zn) were then added to the SF based on the law of garlic nutrient absorption and abundance and deficiency status of soil elements (Figs. 3 and 5). Compared with conventional farmer fertilization, garlic-SF significantly improved sprout and bulb yield as well as nutritional quality (Tables 3 and 4; Fig. 6). However, this considered the dosage of each element but not the raw materials, which also contribute to the quality of the SF.

Soil organic matter content had a significant effect on garlic production. When soil organic matter was insufficient, it reduced about one-third garlic yield (Hong et al. 2024). In this study, there was a significant correlation ($R^2 = 0.804$) between soil organic matter and garlic yield under the conventional management level of farmers (Tables 1 and 3). Low soil organic matter led to the deterioration of soil structure, which was mainly manifested in the destruction of soil aggregate structure, resulting in insufficient mineral elements fixation and further decreased yield (Obalum et al. 2017). Organic fertilizer contains lots of trace elements, which effectively makes up for the lack of trace elements in the soil, but its effect on the improvement of garlic yield in the current season was sometimes not significant. Previous studies have demonstrated that long-term application of organic fertilizer could effectively delay the continuous cropping obstacle of garlic (Wei et al. 2023), but its physiological mechanism was still unclear.

Compared with FP (18:18:18), our garlic special fertilizer not only adjusted the N:P:K ratio (18:10:14), but also added some medium and trace elements (Mg, B, Zn, and Mn) and

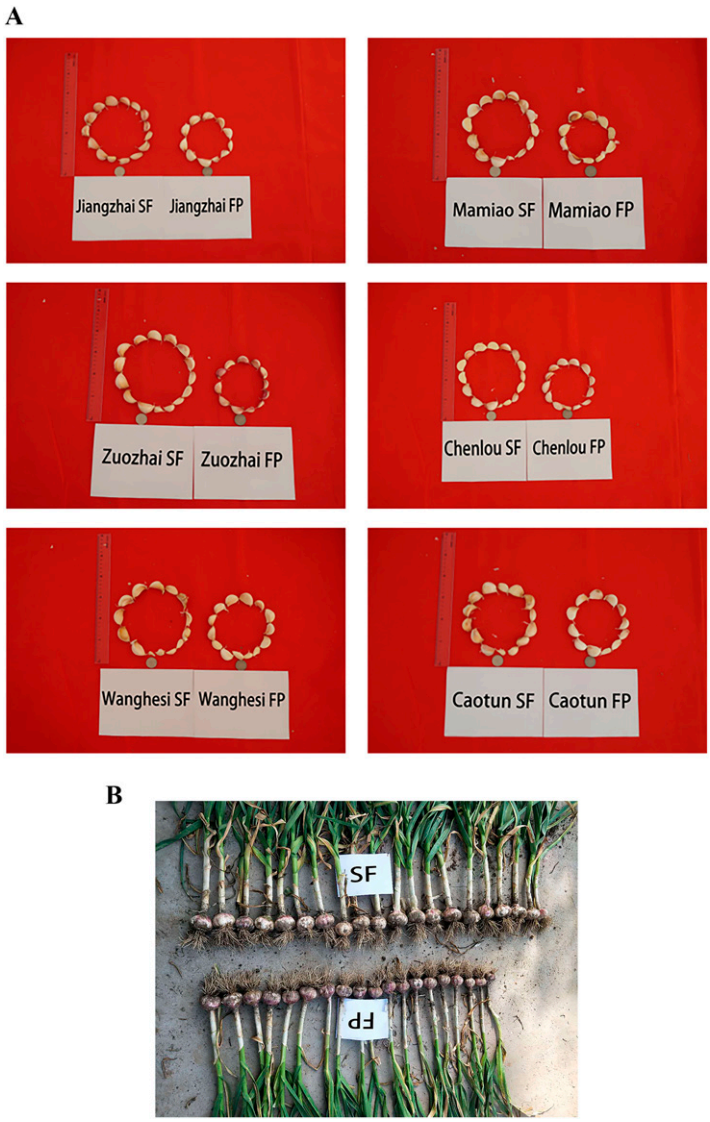

Fig. 6. Mature garlic bulbs (**A**) and groups (**B**) harvested after different fertilizer treatments. (**A**) This circle is made up of all the cloves of a single garlic, which is used to visually express the difference in the number and size of cloves of a single garlic for the two treatments at six experiment locations (Caotun, Jiangzhai, Mamiao, Zuozhai, Wanghesi, Chenlou) during the 2020–21 growing season; (**B**) garlic bulbs were sampled from the Chenlou experiment site during 2020–21. FP = farmer practice; SF = special fertilizer.

adjusted the S:Cl ratio (3:1; Wang et al. 2022). In addition, considering the long growth period of garlic (~210 d), 60% polyurethane controlled-release nitrogen fertilizer was used to instead of available nitrogen fertilizer. The addition of microelements increased the production cost of SFs (557 dollar/t) relative to that of traditional compound fertilizers (529 dollar/t) (Table 3). Although fertilizer prices had increased, increasing production allowed farmers to benefit more (Table 3). Furthermore, the application of SFs improved fertilizer utilization efficiency, thereby reducing risks of water eutrophication (Chien et al. 2009; Dimkpa et al. 2020; Huang et al. 2017; Schröder et al. 2011; Zhang et al. 2011). Gradually replacing balanced fertilizers in the garlic market with garlic-SFs offers economic and environmental benefits. In the future, this SF formulation will be further improved for different types/varieties of garlic. The harvesting organ of garlic can be stem and leaf or sprout or bulb, hence the fertilizer formula, dosage, and application method of garlic need to be adjusted. In addition, as more and more farmers began to apply drip irrigation technology, research on how to fertilize under the drip irrigation mode will be the next focus.

Conclusions

Traditional approaches to garlic fertilization demand large amounts of fertilizer and balanced formulas that result in soil nutrient imbalance, low fertilizer utilization efficiency, and reduced yield and nutritional quality, which decrease farmer profits and risk eutrophication. We developed a garlic-SF 18–10–14 (containing S, Cl, Mn, Fe, and Zn) based on the law of nutrient absorption and soil mineral content. The recommended SF rate was 1500 kg·ha^{-1}, ultimately increasing sprout yield by 12.9% to 30.5%, bulb yield by 11.0% to 33.5%, and net income by 18.2% to 45.6%. The N, P, and K partial productivity also increased by 26.6% to 50.1%, 82.6% to 116.5%, and 54.6% to 83.3%, respectively.

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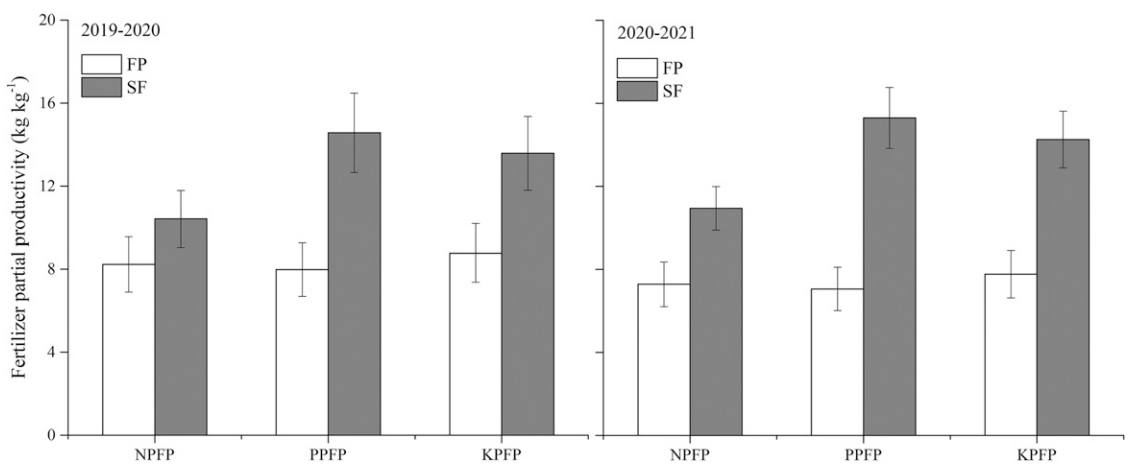

Fig. 7. Effect of special fertilizer on garlic nutrient use efficiency.

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